3.6 Trigonometric equations

The equations containing trigonometric functions of unknown angles are known as trigonometric equations. A solution of trigonometric equation is the value of unknown angle that satisfies the equation. Unless the domain is restricted, the trigonometric equations have infinitely many solutions, a fact due to the periodicity of trigonometric functions. Some of the equations may not have a solution at all.

For example, the equation $\sin\theta = \frac{3}{2}$, does not have solution, since $-1 \leq \sin\theta \leq 1$.

The equation $\sin\theta = 0$ has infinitely many solutions given by $\theta = \pm\pi, \pm 2\pi, \pm 3\pi, \ldots$ and note that these solutions occur periodically.

General Solution

The solution of a trigonometric equation giving all the admissible values obtained with the help of periodicity of a trigonometric function is called the **general solution** of the equation.

Principal Solution

The smallest numerical value of unknown angle satisfying the equation in the interval $[0, 2\pi]$ (or) $[-\pi, \pi]$ is called a **principal solution.** We shall take the interval $[-\pi, \pi]$ for defining the principal solution. Further, in this interval we may have two solutions. Even though both are valid solutions, we take only the numerically smaller one. This helps us to define the principal domain of the trigonometric functions in order to have their inverses.

Principal value of sine function lies in the interval $\left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$ and hence lies in I quadrant or IV quadrant. Principal value of cosine function is in $[0, \pi]$ and hence in I quadrant or II quadrant. Principal value of tangent function is in $\left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$ and hence in I quadrant or IV quadrant.



(i) Trigonometric equations are different from trigonometric identities, since trigonometric identities are true for all admissible values of unknown angle θ. But trigonometric equations are valid only for particular values of unknown angle.

(ii) There is no general method for solving trigonometric equations. However, one may notice that some equations may be factorisable; some equations may be expressed in terms of single function; some equations may be squared.

(iii) To find solutions to trigonometric equations, some times one may go for the technique of squaring both sides. One has to take care as it can also produce false solutions (**extraneous solutions**).

For example, to find solution for $\sin x - \cos x = 1$ in $0 \leq x < 360^\circ$, we do squaring on both sides to get $(\sin x - 1)^2 = \cos^2 x$, which gives $2\sin x(\sin x - 1) = 0$. So, we get $x = 0, \frac{\pi}{2}, \pi$. Clearly $x = 0$ is a false solution. Thus, we have to check for correct solutions, in the squaring process.

(iv) Mostly we write the solutions of trigonometric equations in radians.

Now, we find the solutions to different forms of trigonometrical equations.